

Project Report: Loan Default Risk Analysis & Prediction

Problem Statement

Loan default is one of the major challenges faced by financial institutions. Incorrect lending decisions can lead to significant financial losses and increased credit risk. This project aims to develop a machine learning-based system that predicts whether a loan applicant is likely to default, enabling lenders to make informed and data-driven decisions.

Project Overview

This project implements an end-to-end machine learning pipeline for predicting loan default risk using borrower demographic, financial, and loan-related information. Multiple classification models were developed and compared, and the best-performing model was deployed using a Streamlit web application that allows users to interactively predict the probability of loan default.

Dataset Summary

The dataset consists of **255,347** loan records with **18 variables** describing borrower characteristics and loan details.

Predictor Variables

1. Loan id
2. Age
3. Income
4. Loan Amount
5. Credit Score
6. Months Employed
7. Number of Credit Lines
8. Interest Rate
9. Loan Term
10. Debt-to-Income Ratio (DTI)
11. Education
12. Employment Type
13. Marital Status
14. Mortgage Status
15. Number of Dependents
16. Loan Purpose
17. Co-signer Availability

Target Variable

18. Default

1 – Loan Default

0 – No Default

Data Preprocessing

The following preprocessing steps were performed before model training:

- Removed non-informative identifier column (LoanID)
- One-Hot Encoding of categorical variables
- Train-Test Split (80:20)
- Standardization using StandardScaler
- Addressed class imbalance using SMOTE on the training data
- Saved preprocessing objects (Scaler and Feature Names) for deployment

Exploratory Data Analysis (EDA Highlights)

The exploratory analysis revealed several important insights:

- The dataset exhibited **class imbalance**, with non-default cases significantly outnumbering default cases.
- Borrowers with **lower credit scores** showed a greater tendency to default.
- Higher **Debt-to-Income (DTI) ratios** were associated with increased default risk.
- Employment status and education level demonstrated noticeable relationships with repayment behaviour.
- Correlation analysis identified important numerical variables influencing default prediction.

Model Development

Four supervised machine learning algorithms were implemented and evaluated:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest

Hyperparameter tuning was performed using **GridSearchCV** where appropriate.

Although **Random Forest** achieved the highest Accuracy and ROC-AUC, **Logistic Regression** produced the highest **F1-score**, making it the preferred model for deployment due to the imbalanced nature of the dataset.

Model Performance

Model	Accuracy	F1 Score
Logistic Regression	81.17%	24.41%
KNN	80.21%	18.50%
Decision Tree	77.47%	23.19%
Random Forest	85.41%	17.62%

The final deployed model was **Logistic Regression**.

Evaluation Metrics

The models were evaluated using multiple performance metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

Since the dataset was imbalanced, F1-score was considered the primary metric for selecting the final model.

Streamlit Application

A user-friendly **Streamlit** application was developed to demonstrate the trained model.

The application allows users to:

- Enter borrower information.
- Predict the likelihood of loan default.
- Display the probability of default.
- Provide a simple risk assessment (Low Risk / High Risk).

The application uses the saved Logistic Regression model together with the preprocessing objects to generate real-time predictions.

Challenges

Several challenges were encountered during the project:

- Handling significant class imbalance in the dataset.
- Selecting an evaluation metric appropriate for imbalanced classification.
- Balancing model interpretability with predictive performance.
- Ensuring that the preprocessing pipeline remained consistent during deployment.

Impact

This project demonstrates how machine learning can support financial institutions by:

- Assisting in loan approval decisions.
- Identifying high-risk applicants.
- Reducing financial losses due to defaults.
- Supporting risk analysts with data-driven insights.
- Serving as an educational prototype for credit risk modelling.

Tech Stack

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Imbalanced-learn (SMOTE)
- Joblib
- Streamlit
- Jupyter Notebook

Future Work

Potential enhancements for this project include:

- Implementing advanced ensemble methods such as **XGBoost**, **LightGBM**, and **CatBoost**.
- Incorporating Explainable AI techniques such as **SHAP** and **LIME** for model interpretability.
- Integrating real-world banking datasets.
- Deploying the application on Streamlit Community Cloud.
- Developing an interactive dashboard with visual analytics for financial institutions.